Lab Exercise - 3

[Based on Python Data Types (String, List , Conditional statements and Loops]

Q1: Write a Python program to sum all the items in a list.

Q2: Write a Python program to get the largest number from a list.

Q3: Write a Python program to get the smallest number from a list.

Q4: Write a Python program to display the first and last colors from the following list.

color_list = ["Red","Green","White" ,"Black"]

Q5. Write a program that prints minimum and maximum of five numbers entered by the user.

Q6: write a Program to check if a string is palindrome or not

Q7: Given a nested list. Write a python program to extend it with adding sub list ["h", "i", "j"] in a such a way that it will look like the following list

Given List:

list1 = ["a", "b", ["c", ["d", "e", ["f", "g"], "k"], "l"], "m", "n"]

Sub List to be added = ["h", "i", "j"]

Expected output:

['a', 'b', ['c', ['d', 'e', ['f', 'g', 'h', 'i', 'j'], 'k'], 'l'], 'm', 'n']

Q8: Write a python program for Given a Python list, to find value 20 in the list, and if it is present, replace it with 200. Only update the first occurrence of a value

list1 = [5, 10, 15, 20, 25, 50, 20]

Expected output:

list1 = [5, 10, 15, 200, 25, 50, 20]

Q9. Write a program to generate the sequence: –5, 10, –15, 20, –25….. upto n, where n is an integer input by the user.

Q10. Write a program to find the sum of 1+ 1/8 + 1/27.....$1/n^3$, where n is the number input by the user.

Q11. Write a program to find the sum of digits of an integer number, input by the user.

Q12: Write a python program to display ascii characters from 65 to 90

```
65  =  A
66  =  B
67  =  C
68  =  D
69  =  E
70  =  F
71  =  G
72  =  H
73  =  I
74  =  J
75  =  K
76  =  L
77  =  M
78  =  N
79  =  O
80  =  P
81  =  Q
82  =  R
83  =  S
84  =  T
85  =  U
86  =  V
87  =  W
88  =  X
89  =  Y
90  =  Z
```

Q13: Display ascii characters from 48 to 57.

```
48  =  0
49  =  1
50  =  2
51  =  3
52  =  4
53  =  5
54  =  6
55  =  7
56  =  8
57  =  9
```

Q14: Display the following output with the help of Ascii character.

```
97  =  a
98  =  b
99  =  c
100  =  d
101  =  e
102  =  f
103  =  g
104  =  h
105  =  i
106  =  j
107  =  k
108  =  l
109  =  m
110  =  n
111  =  o
112  =  p
113  =  q
114  =  r
115  =  s
116  =  t
117  =  u
118  =  v
119  =  w
120  =  x
```

Q15: Given a Python list. Write a python program to turn every item of a list into its square List1 = [1, 2, 3, 4, 5, 6, 7]

Expected output:

[1, 4, 9, 16, 25, 36, 49]

Q16: write a Program to sort alphabetically the words form a string provided by the user.

[You can use split() method to split string into a list of words.]

Q17: write a Program to Remove Punctuations from a String provided by the user. [Hint: use punctuation attribute of **string module** to get all punctuations (i.e. **!"#$%&\'()*+,-./:;<=>?@[\\]^_`{|}~**)]

Q18. Write a menu driven program to develop SMIS that has options to

• accept the marks of the student in five major subjects in Class and display the same.

• calculate the sum of the marks of all subjects. Divide the total marks by number of subjects (i.e. 5), calculate percentage = total marks/5 and display the percentage.

• Find the grade of the student as per the following criteria:

Criteria	Grade
percentage > 85	A
percentage < 85 && percentage >= 75	B
percentage < 75 && percentage >= 50	C
percentage > 30 && percentage <= 50	D
percentage <30	Reappear

Sample output:

```
--- Student Management Information System (SMIS) ---
1. Enter marks of 5 major subjects
2. Calculate Total and Percentage
3. Determine Grade
4. Exit
Select an option (1-4): 2
Error: No marks found. Please select Option 1 first.

--- Student Management Information System (SMIS) ---
1. Enter marks of 5 major subjects
2. Calculate Total and Percentage
3. Determine Grade
4. Exit
Select an option (1-4): 3
Error: Please enter marks and calculate percentage first.

--- Student Management Information System (SMIS) ---
1. Enter marks of 5 major subjects
2. Calculate Total and Percentage
3. Determine Grade
4. Exit
Select an option (1-4): 1
Enter marks for Subject 1: 78
Enter marks for Subject 2: 89
Enter marks for Subject 3: 96
Enter marks for Subject 4: 89
Enter marks for Subject 5: 78
Marks recorded successfully: [78.0, 89.0, 96.0, 89.0, 78.0]

--- Student Management Information System (SMIS) ---
1. Enter marks of 5 major subjects
2. Calculate Total and Percentage
3. Determine Grade
4. Exit
Select an option (1-4): 2
Total Marks: 430.0
Percentage: 86.0%

--- Student Management Information System (SMIS) ---
1. Enter marks of 5 major subjects
2. Calculate Total and Percentage
3. Determine Grade
4. Exit
Select an option (1-4): 3
Calculated Grade: A

--- Student Management Information System (SMIS) ---
1. Enter marks of 5 major subjects
2. Calculate Total and Percentage
3. Determine Grade
4. Exit
Select an option (1-4): 4
Exiting SMIS. Goodbye!
```

[Nested Loops]

Q19: Write a python program to print the Following:

1

2 1

3 2 1

Q20: WAP to print the following asterisk pattern:

```
*
* *
* * *
* * * *
* * * * *
* * * * * *
```

Q21: WAP to create a function traiangle to print the following asterisk triangle pattern:

```
*
* *
* * *
* * * *
* * *
* *
*
```

Q22: Write a python program to print following multiplication table on the screen

1 |	1	2	3	4	5	6	7	8	9	10
2 |	2	4	6	8	10	12	14	16	18	20
3 |	3	6	9	12	15	18	21	24	27	30
4 |	4	8	12	16	20	24	28	32	36	40
5 |	5	10	15	20	25	30	35	40	45	50
6 |	6	12	18	24	30	36	42	48	54	60
7 |	7	14	21	28	35	42	49	56	63	70
8 |	8	16	24	32	40	48	56	64	72	80
9 |	9	18	27	36	45	54	63	72	81	90
10 |	10	20	30	40	50	60	70	80	90	100